

BioHackathon series:

[Biohackathon SWAT4HCLS 2026](#)

AmsterdamUMC

[OpenMRS](#)

Submitted: 25 Mar 2026

License:

Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC-BY](#)).

Published by [BioHackrXiv.org](#)

SWAT4HCLS26 report: OpenMRS as a reference implementation platform for Electronic Health Records and Clinical Decision Support

Andra Waagmeester ¹, **Stephanie “Ace” Medlock**¹, and **Claude Nanjo**²

¹ Department of Medical Informatics, AmsterdamUMC, Location AMC Amsterdam, the Netherlands

² University of Utah

Introduction

The broad goal for the OpenMRS table is: What steps are needed to use OpenMRS as a resource for reference implementations?

This can be divided into 4 levels:

1. Use OpenMRS as a data source, using its available APIs This is the easiest one. I will set up an OpenMRS instance that can be used, and provide the URL, some logins, and some instructions. The goal for anyone working on this will simply be to work on whatever demo they want to work on, using the APIs and the fake data provided by this system.
2. Use OpenMRS running locally, so that you can directly access its database The instructions provided by the OpenMRS community for the newest version (O3) are actually not bad, and I have some detailed notes about how to do this on Linux that I can share with the group. I've never done it on Windows, but it can't be that hard. The goal for anyone working on this will be to come up with instructions for installing the system on Windows, and querying the database (e.g. from Python). This is probably what Claude Nanjo will need to start with.
3. Add a module to OpenMRS This is the first step toward making modifications to OpenMRS itself. The OpenMRS community provides some “instructions for developers”. The goal for anyone working on this will be to follow those instructions (and improve them where needed) with the goal of adding a simple module (I would suggest “add a clickable URL to the patient summary screen” as a good objective for MyFirstModule). This is what will be needed for Claude's goal, if OpenMRS does not already have a module that supports data entry for the data he needs for his idea.
4. Build OpenMRS from source This is what we'll be working on. This is what you need if you want to change something in OpenMRS, rather than just adding something to it. The goal will be to come up with a set of instructions for how it can be built from source... preferably in an executable format (e.g. a Makefile).

Meeting information

If you want to submit a preprint to BioHackrXiv, first check if your meeting is registered. You can find a list of meetings [here](#). If your meeting is missing, please contact your meeting organizers. The above list also provides information on the YAML fields with information about the meeting.

The following fields need to be given:

`biohackathon_name:` "BioHackathon Europe 2023"

`biohackathon_url:` "<https://biohackathon-europe.org/>"



```
biohackathon_location: "Barcelona, Spain, 2023"
group: Project 26
git_url: https://github.com/yourOrganization/your_report_repo
```

The [BioHackrXiv meeting pages](#) provide content to use for the first three fields. The `git_url` field must have the link to the GitHub repository with your preprint (draft).

Author information

Information about the authors is given in the [YAML](#) format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information is encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

Author identifiers

Ideally, authors provide their [ORCID](#) identifier. For affiliations, it is added with the `orcid` field. So, an author record would look like this:

```
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
```

Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the [Research Organization Registry](#) (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror` field:

```
affiliations:
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role` field. However, the authors are responsible for selecting the right terms from [CRediT](#). An example looks like this:

```
authors:
  - name: First Author
    affiliation: 1
```



```
orcid: 0000-0000-0000-0000
role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```
authors:
- name: First Author
  affiliation: 1
  role: Writing - original draft
- name: Last Author
  orcid: 0000-0000-0000-0000
  affiliation: 2
  role: Conceptualization, Writing - review & editing
affiliations:
- name: First Affiliation
  index: 1
- name: ELIXIR Europe
  ror: 044rwnt51
  index: 2
```

Formatting

This document use Markdown and you can look at [this tutorial](#).

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

Table: Note that table caption is automatically numbered and should be given before the table itself.

```
| Header 1 | Header 2 |
| ----- | ----- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```



This gives:



Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use [CiTO](#) annotations, as explained in [this BioHackathon Europe 2021 write up](#) and [this CiTO Pilot](#). Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethod In:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- `citesAsDataSource`: when you point the reader to a source of data which may explain a claim
- `usesDataFrom`: when you reuse somehow (and elaborate on) the data in the cited entity
- `usesMethodIn`
- `citesAsAuthority`
- `citesAsEvidence`
- `citesAsPotentialSolution`
- `citesAsRecommendedReading`
- `citesAsRelated`
- `citesAsSourceDocument`
- `citesForInformation`
- `confirms`
- `documents`
- `providesDataFor`
- `obtainsSupportFrom`

- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

...

Acknowledgements

...

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, 32(14), 2236–2238. <https://doi.org/10.1093/bioinformatics/btw214> [cito:citesAsAuthority]

Author contributions

Andra Waagmeester: Writing – original draft
Stephanie “Ace” Medlock: Writing – original draft
Claude Nanjo: Writing – original draft